

Placement Empowerment Program
Cloud Computing and DevOps Centre

Set Up IAM Roles and Permissions

Create an IAM role on your cloud platform. Assign the role to your VM to restrict/allow specific actions.

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Introduction and Overview

This Proof of Concept (PoC) aims to demonstrate the process of assigning an IAM role to an AWS EC2 instance. IAM roles are used to grant permissions to AWS services without needing to store credentials manually. By assigning an IAM role, an EC2 instance can securely access AWS resources based on the role's policies.

Objective

- Verify if an IAM role is assigned to an EC2 instance.
- Assign an appropriate IAM role to an instance if not already assigned.
- Retrieve and validate IAM role credentials.

Importance

IAM roles enhance security and compliance by eliminating the need to hardcode credentials in applications. They provide controlled access to AWS services and enable secure authentication for EC2 instances. Implementing IAM roles ensures a scalable and manageable approach to permissions in cloud environments.

Steps for Execution

1. **Check IAM Role Assignment:** Access the AWS EC2 instance and verify if an IAM role is assigned by querying instance metadata.

EC2-Instance-Access-Role

Info

Delete

Allows EC2 instances to call AWS services on your behalf.

Summary

Edit

Creation date

February 23, 2025, 22:44 (UTC+05:30)

ARN

arn:aws:iam::116981790509:role/EC2-Instance-Access-Role

Instance profile ARN

arn:aws:iam::116981790509:instance-profile/EC2-Instance-Access-Role

Last activity

-

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1)

Info

Refresh

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type

All types

< 1 >

Settings

☐

Policy name

☐

AdministratorAccess

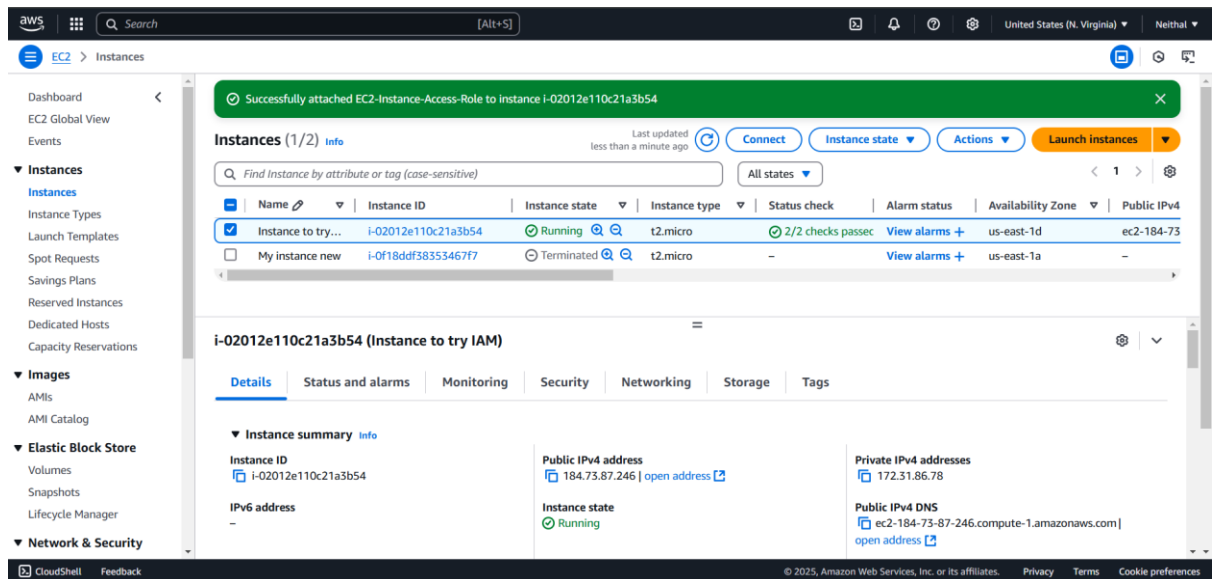
Type

AWS managed - job function

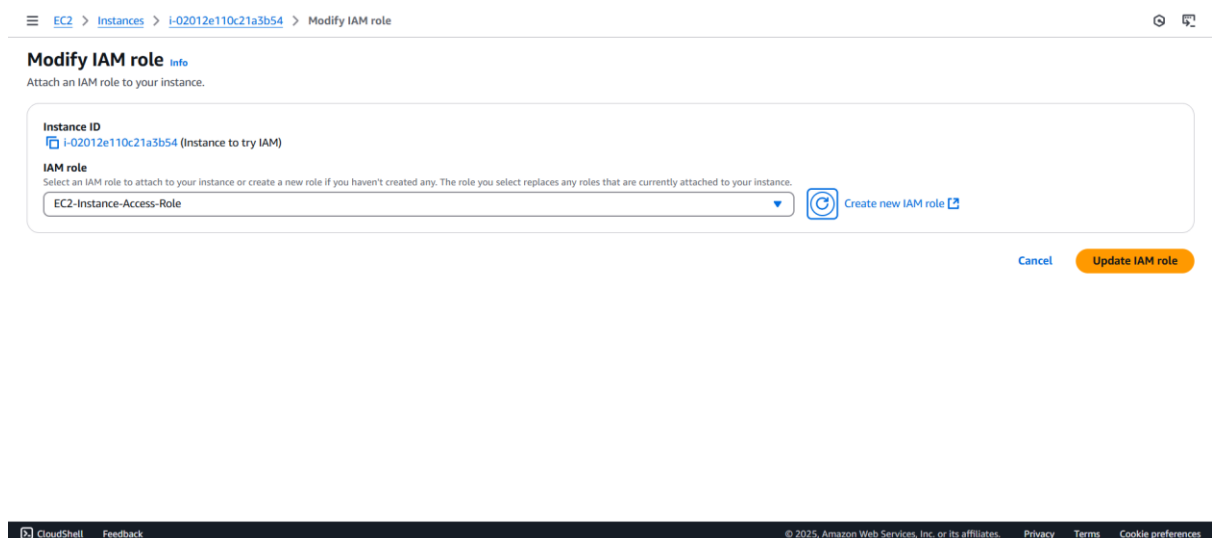
Attached entities

1

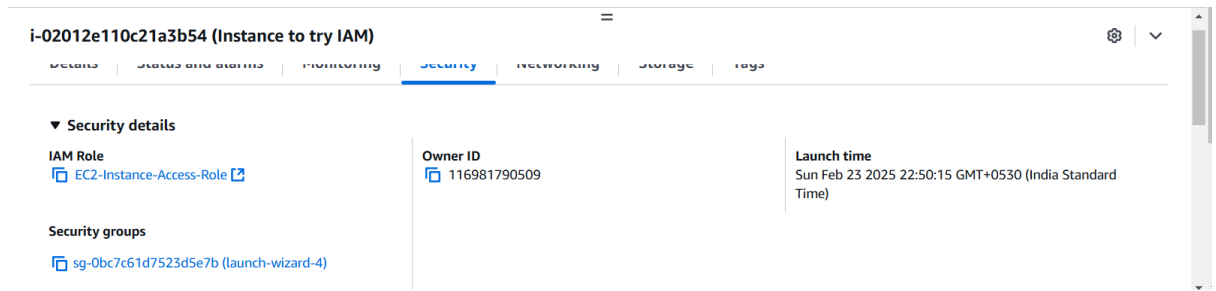
2. **Navigate to AWS EC2 Dashboard:** Open the AWS Management Console and go to the EC2 service to check the instance details.



3. **Modify IAM Role:** If no IAM role is attached, modify the instance settings to attach an existing IAM role with appropriate permissions.
4. **Confirm IAM Role Attachment:** Recheck the instance metadata after role assignment to ensure it is applied successfully.



5. **Validate Credentials:** Access the IAM role details and verify that the instance can retrieve temporary security credentials, ensuring that permissions are correctly assigned.



Conclusion

Successfully assigning an IAM role to an EC2 instance ensures secure and controlled access to AWS services without manual credential management. This process enhances security, simplifies permission management, and follows best practices for cloud security.

What I Learned

- How to check if an IAM role is assigned to an EC2 instance.
- The process of attaching and modifying IAM roles in the AWS Management Console.
- The significance of IAM roles in improving cloud security and permissions management.
- How to validate the assigned IAM role and its permissions.
- Best practices for secure authentication in AWS environments.